hdoc day 43 by trishil chauhan

Aim

Write a menu-driven C program using separate user-defined functions to find the longest word in a sentence and check whether one string is a rotation of another.

Complete C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define SIZE 500

void readLine(char str[], size_t size) {
    if (fgets(str, (int)size, stdin) != NULL) {
        str[strcspn(str, "\n")] = '\0';
    }
}

void findLongestWord(const char sentence[], char longest[]) {
    int i = 0, start = 0, maxLen = 0;
    int length = (int)strlen(sentence);
    longest[0] = '\0';

    while (i <= length) {
        if (i < length && !isspace((unsigned char)sentence[i])) {
            i++;
        } else {
            int wordLen = i - start;
            if (wordLen > maxLen) {
                maxLen = wordLen;
                memcpy(longest, sentence + start, (size_t)wordLen);
                longest[wordLen] = '\0';
            }
            i++;
            start = i;
        }
    }
}

int isRotation(const char first[], const char second[]) {
    size_t len = strlen(first);
    char doubled[2 * SIZE];
    if (len != strlen(second)) return 0;
    if (len == 0) return 1;
    strcpy(doubled, first);
    strcat(doubled, first);
    return strstr(doubled, second) != NULL;
}

int main(void) {
    char sentence[SIZE], longest[SIZE];
    char first[SIZE], second[SIZE];
    int choice;

    do {

        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Find longest word in a sentence\n");
        printf("2. Check string rotation\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
```

```
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input. Exiting.\n");
            break;
        }
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }

        switch (choice) {
            case 1:
                printf("Enter a sentence: ");
                readLine(sentence, SIZE);
                findLongestWord(sentence, longest);
                if (longest[0] == '\0')
                    printf("No word found.\n");
                else
                    printf("Longest word: %s\n", longest);
                break;
            case 2:
                printf("Enter first string: ");
                readLine(first, SIZE);
                printf("Enter second string: ");
                readLine(second, SIZE);
                if (isRotation(first, second))
                    printf("The strings are rotations.\n");
                else
                    printf("The strings are not rotations.\n");
                break;
            case 3:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice. Try again.\n");
        }
    } while (choice != 3);

    return 0;
}
```

Sample Output

```
--- STRING OPERATIONS MENU ---
1. Find longest word in a sentence
2. Check string rotation
3. Exit
Enter your choice: 1
Enter a sentence: Programming is interesting
Longest word: Programming

Enter your choice: 2
Enter first string: abcde
Enter second string: cdeab
The strings are rotations.

Enter your choice: 3
Exiting program.
```

Notes

The longest word is defined as the longest whitespace-separated token; ties return the first occurrence. Rotation checking is case-sensitive and uses the doubled-first-string technique.